

Participation and Spending in the Medicare Shared Savings Program

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Outline

Introduction

Institutional Setting

Data and Reduced Form Evidence

Structural Model and Estimation Results

Counterfactual Analysis

Motivation

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- ▶ Policy relevance: CBO report on benchmarking issues

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 - (ii) Spending ↓ by additional 1.13 pp (\$5.085 bln) relative to MSSP.

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► Pay-for-Performance:

- framework can be applied to other P4P programs
- benchmarking rules in a dynamic setting
(Gaynor, Rebitzer, Taylor, 2004; Ho and Pakes, 2014; Eliason et al., 2018; Zhang et al. 2018; Einav, Finkelstein et al., 2021;)

Overview of this Presentation

- ▶ Institutional Setting
- ▶ Data and Reduced Form Evidence
- ▶ Structural Model and Estimation Results
- ▶ Counterfactual Analysis

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- ▶ The bonus is called shared savings

The Medicare Shared Savings Program

- Bonus = Shared Savings (SS) [» details](#)

$$SS_{it} = \begin{cases} s \cdot (b_{it} - y_{it}^{MSSP}) & \text{if } y_{it}^{MSSP} < b_{it} \\ 0 & \text{otherwise} \end{cases}$$

where $s \in [0, 1]$ is the shared-savings rate (e.g. $s = 0.5$).

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- Benchmark

$$b_{it} = (1 - \lambda) \underbrace{b_{it}^h}_{\text{historical rebased benchmark}} + \lambda \underbrace{y_{it}^R}_{\text{average regional spending}}$$

where $\lambda \in [0, 1]$ is the regional weight and

$$b_{it}^h = (y_{it-1} + y_{it-2} + y_{it-3})/3$$

is the average ACO's past spending.

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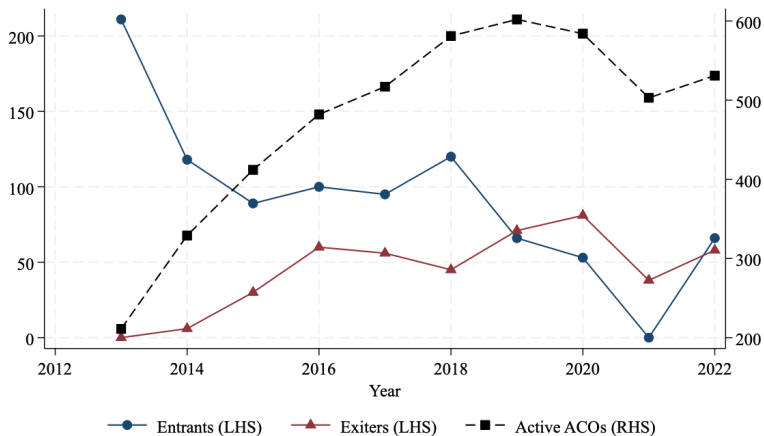
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Participation in the MSSP



Data Source: CMS - MSSP Performance Data 2013-2022

Figure 1: Entry, Exit and Active ACOs from 2013 to 2022

Ratchet Effect - Reduced Form Evidence

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- Estimate

$$\underbrace{\text{Savings Rate}_{it}}_{(b_{it} - y_{it})/b_{it}} = \alpha_i + \lambda_t + \beta \cdot \text{Non-RY}_{it} + \mathbf{X}_{it}'\delta + \varepsilon_{it}$$

Ratchet Effect - Reduced Form Evidence

Table 1: Ratchet Effect: Two Way Fixed Effects Regressions

	OLS	
	Full sample	Entry year < 2019
Non-Rebasing Year	0.943*** (0.210)	0.907*** (0.227)
Observations	4,519	4,133
R-squared	0.180	0.186
Mean	2.324	2.235
SD	4.173	4.207

Notes: (i) Dependent variable: CMS savings rate $s_t = (b_t - y_t)/b_t$. (ii) Year FE and ACO FE are included. (iii) Standard errors clustered by ACO.

Adverse Selection - Reduced Form Evidence

► Recall:

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► Estimate a logistic regression

$$\Pr(\text{Exit}_{it} = 1) = \Lambda\left(\lambda_t + \beta_1 \cdot \text{Regional Adj}_{it} + \beta_2 \cdot \text{Positive Adj}_{it} + X'_{it}\delta\right)$$

Adverse Selection - Reduced Form Evidence

Table 2: Selection on Level: Fixed Effects Logistic Regressions

	(1)	(2)	(3)	(4)
Regional Adj	3.641** (1.159)	3.693*** (0.739)	3.735* (2.615)	3.773*** (0.740)
Positive Adj	0.257*** (0.077)	0.266*** (0.036)	0.299** (0.146)	0.296*** (0.037)
McFadden R ²	0.157	0.155	0.141	0.139
AP FE	Yes	No	Yes	No
Year FE	Yes	Yes	No	No

Dependent Variable: Indicator for Exit from MSSP

Coefficients represents the odds ratio of Exit = 1

AP FE stands for agreement period (cycles) fixed effects

Standard errors clustered by ACO

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where γ_{it} is an unknown ACO specific parameter.

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- ▶ Profits per-capita

$$\pi_{it}^{MSSP}(e_{it}) = \underbrace{SS(y_{it}^{MSSP}, b_{it})}_{\text{shared savings}} - \underbrace{C(e_{it})}_{\text{variable costs}} - \underbrace{F_i}_{\text{fixed costs}}$$

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- ▶ Effort e_{it} is an intertemporal choice. ▶ euler equation

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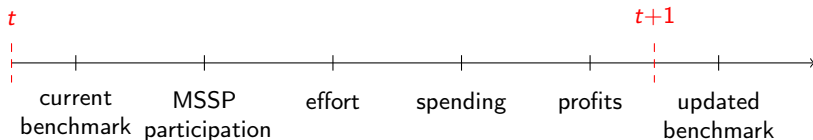
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ACOs Primitives and Savings

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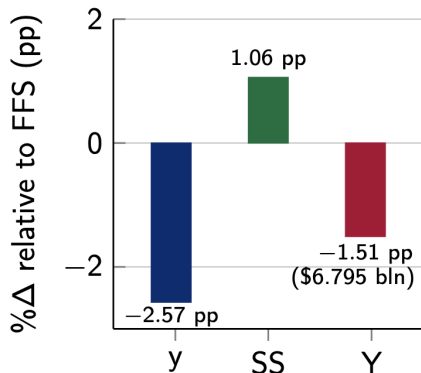
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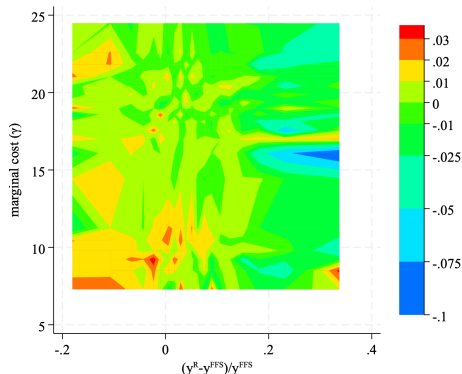
- ▶ How large does effort e_{it} need to be?

$$\underbrace{(1 - s)e_{it}}_{\text{effort driven Medicare savings}} > \underbrace{s(b_{it} - y_{it}^{FFS})}_{\text{free shared savings paid to ACO}}$$

Does MSSP generate savings relative to FFS?



Net Spending (y), Shared Savings (SS),
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x-axis = adverse selection incentive

y-axis = marginal cost of effort

z-axis = $(Y - y^{FFS})/y^{FFS}$

Outline

Introduction

Institutional Setting

Data and Reduced Form Evidence

Structural Model and Estimation Results

Counterfactual Analysis

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Counterfactual without Ratchet Effect

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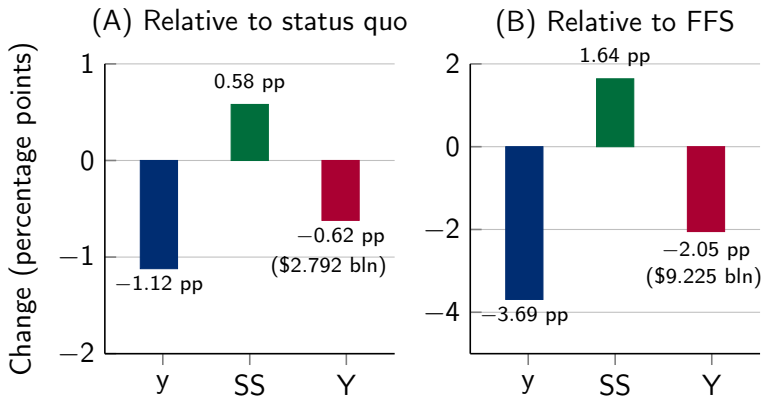
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$$b_{it} = \underbrace{\frac{y_{i-1} + y_{i-2} + y_{i-3}}{3}}_{\substack{\text{average spending} \\ \text{before entry} \\ \text{(initial benchmark)}}} + \underbrace{\lambda (y_t^R - b_t^h)}_{\text{regional adjustment}}$$

No Rebasement vs. status quo and FFS



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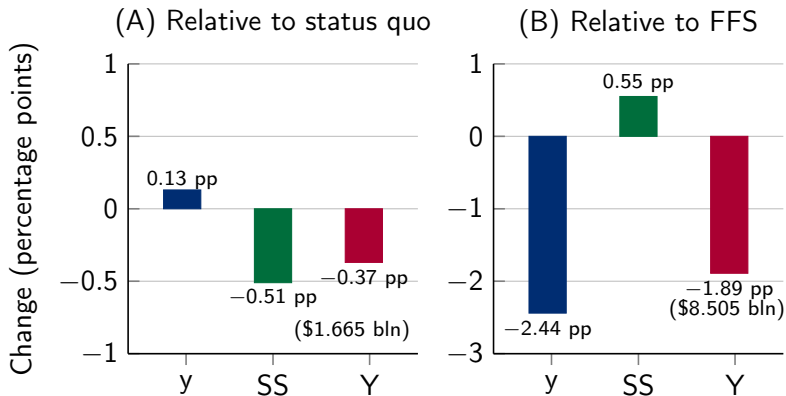
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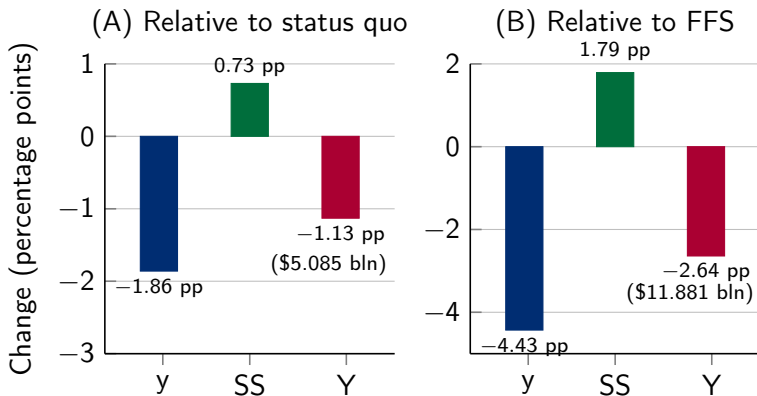
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Thank You!

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MSSP Contract

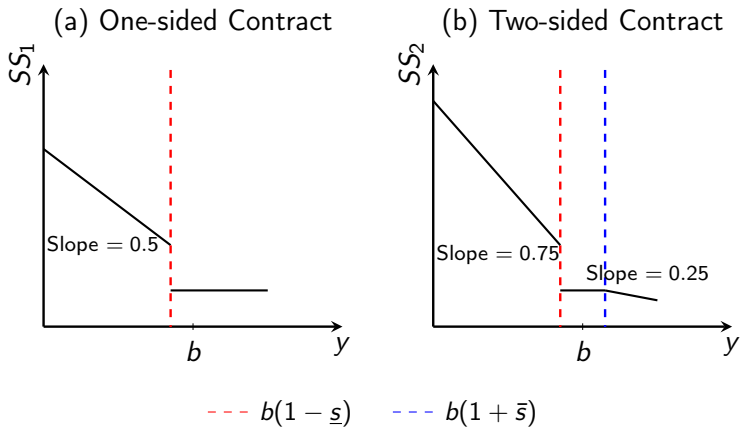
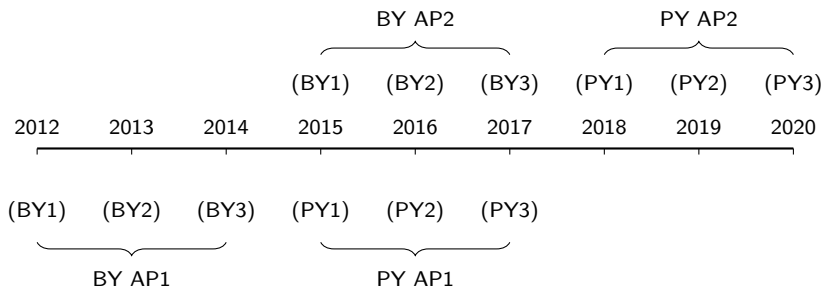


Figure 2: Shared Savings Rewards

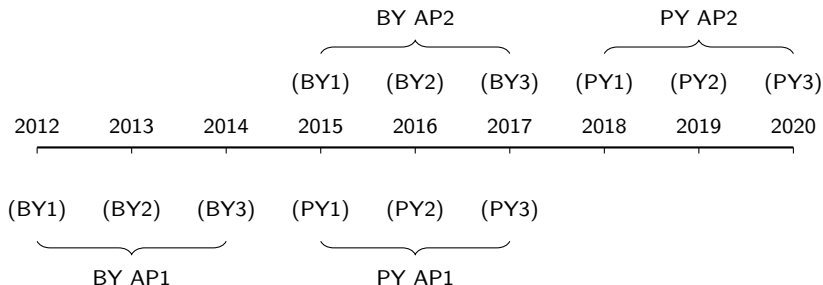
Benchmark over time

- Consider an ACO that joined in 2015 [» back](#)



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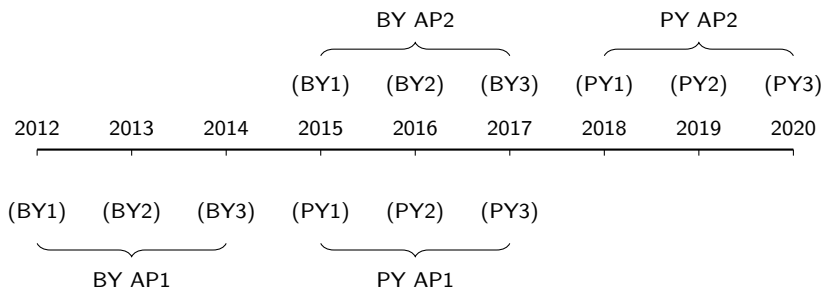


The 1st AP **historical** benchmark is

$$b_{AP1}^h = \underbrace{0.6y_{2014} + 0.3y_{2013} + 0.1y_{2012}}_{\text{average spending prior to joining MSSP}}$$

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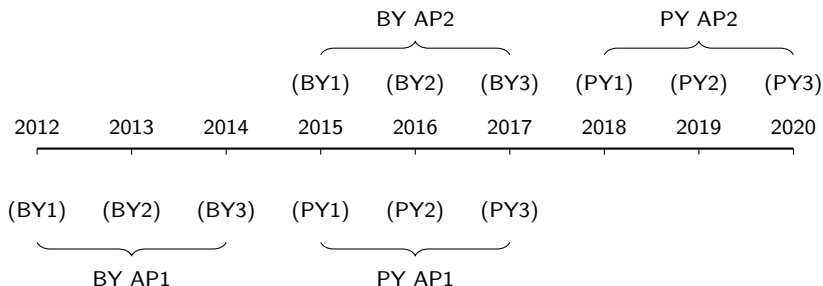


In the 2nd AP benchmark is **rebased**

$$b_{AP_2}^h = \underbrace{\frac{1}{3}y_{2015} + \frac{1}{3}y_{2016} + \frac{1}{3}y_{2017}}_{\text{average spending in the 1st AP}}$$

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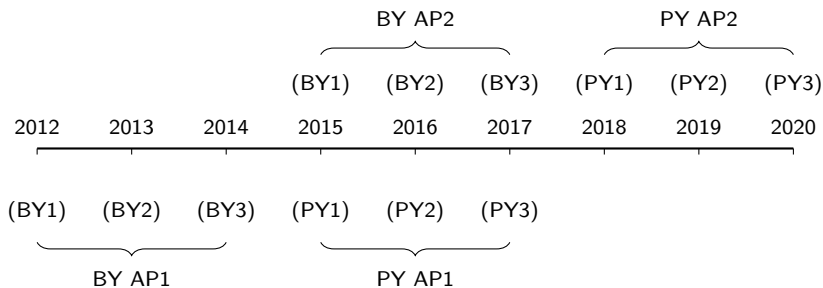


and **regionalized**

$$b_{AP2}^r = (1 - \lambda) \underbrace{b_{AP2}^h}_{\text{rebased benchmark}} + \lambda \underbrace{y^R}_{\text{regional spending}}$$

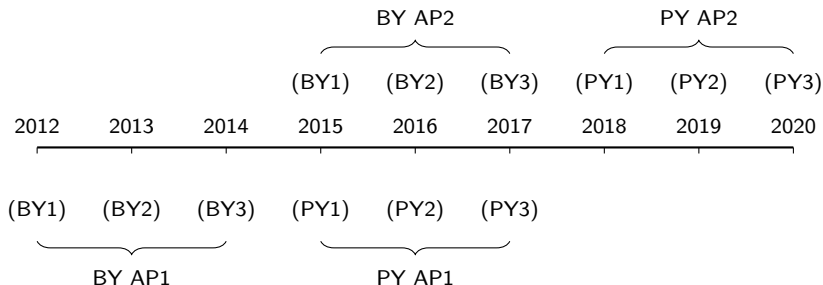
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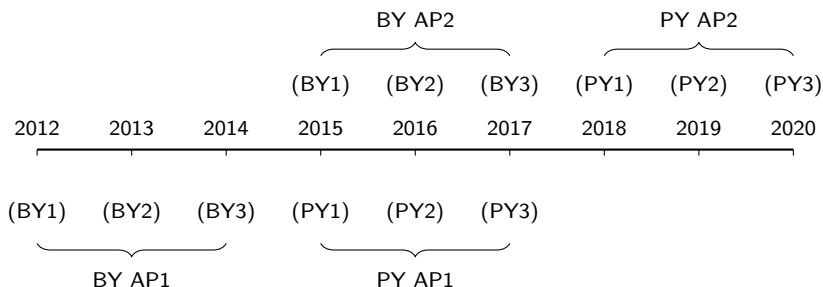


The 1st AP **historical** benchmark is

$$b_{AP1}^h = \underbrace{0.6y_{2014} + 0.3y_{2013} + 0.1y_{2012}}_{\text{average spending prior to joining MSSP}}$$

Benchmark over time

- Consider an ACO that joined in 2015 [▶ back](#)

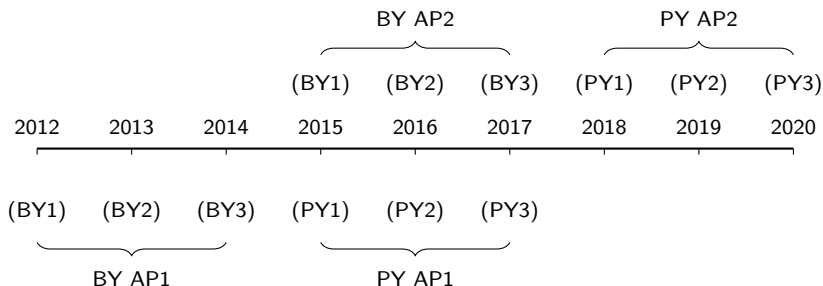


In the 2nd AP benchmark is **rebased**

$$b_{AP_2}^h = \underbrace{\frac{1}{3}y_{2015} + \frac{1}{3}y_{2016} + \frac{1}{3}y_{2017}}_{\text{average spending in the 1st AP}}$$

Benchmark over time

- Consider an ACO that joined in 2015 [▶ back](#)

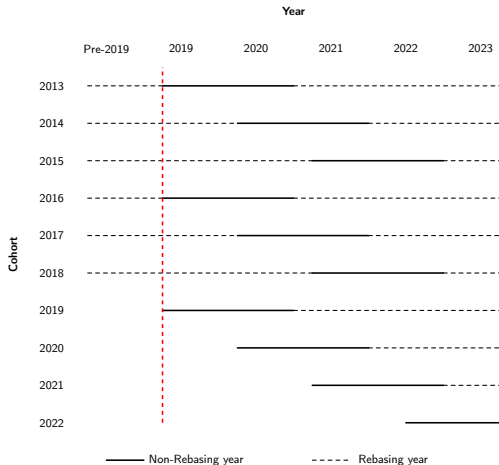


and **regionalized**

$$b_{AP2}^r = (1 - \lambda) \underbrace{b_{AP2}^h}_{\text{rebased benchmark}} + \lambda \underbrace{y^R}_{\text{regional spending}}$$

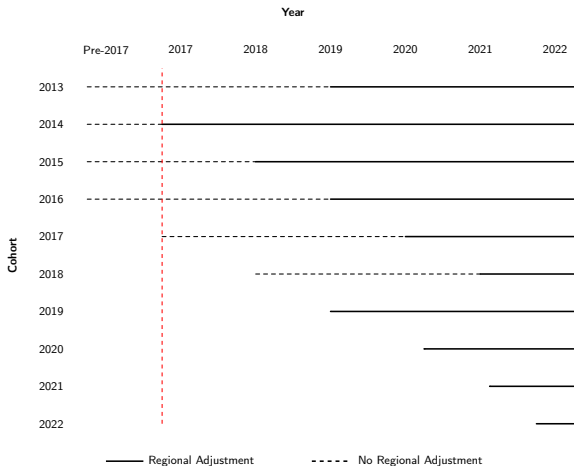
Non-Rebasing Years

Policy: starting from 2019, cycles last 5 years and the first two years do not affect future benchmark. [» back](#)



Regional Adjustment

Policy: regional adjustment ($\lambda > 0$) was introduced at different time across ACO cohorts. [» back](#)



FFS spending

ACO i in year t faces FFS spending y_{it}^{FFS}

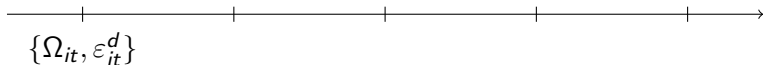
$$\ln y_{it}^{FFS} = \ln y_{it-1}^{FFS} + \ln \text{tf}_{it} + \ln \text{rr}_{it} + u_{it}$$

- ▶ tf_{it} is the national trend and regional FFS trends factor,
- ▶ rr_{it} is the risk ratio,
- ▶ spending shock u_{it} with ϵ_{it} i.i.d. from $N(0, \sigma_{it}^2)$.

▶ back

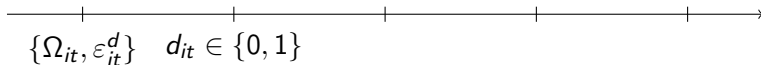
ACOs' Dynamic Problem

► Timing of the model



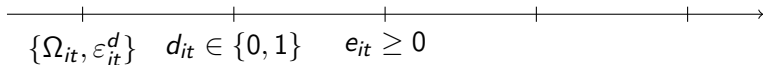
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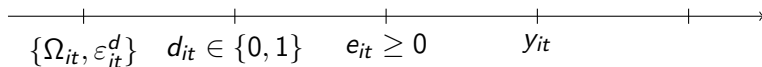
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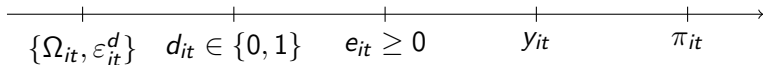
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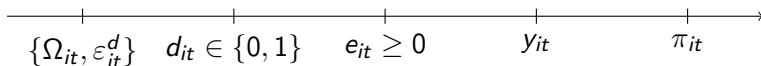
ACOs' Dynamic Problem

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ACOs' Dynamic Problem

- ▶ Timing of the model



- ▶ The MSSP participation condition is given by

$$d_{it} = 1 \iff v_{it}^M(\Omega_{it}) + \varepsilon_{it}^M > v_{it}^F(\Omega_{it}) + \varepsilon_{it}^F$$

where v_{it}^M is the Choice Specific Value Function of MSSP

$$v_{it}^M(\Omega_{it}) + \varepsilon_{it}^M = \max_{e_{it}} \left\{ \pi_{it}^M(e_{it}) + \delta \mathbb{E} V_{it+1}(\Omega_{it+1}) \right\} + \varepsilon_{it}^M$$

where ε_{it}^M is a TIEV choice-specific shock.

Euler Equation

The optimal level of effort at time t satisfies the FOC

$$\frac{\partial \pi_t^M}{\partial e_t} = \begin{cases} 0 & \text{if } t \text{ is non benchmark year} \\ \delta \Pr(d_{t+1} = 1 \mid \tilde{y}_{t+1}, b_{t+1}) \frac{\partial v_{t+1}^M}{\partial \tilde{y}_{t+1}} \frac{\partial \tilde{y}_{t+1}}{\partial y_t} \frac{\partial y_t}{\partial e_t} & \text{otherwise} \end{cases}$$

where $\frac{\partial y_t}{\partial e_t} = -1$

$$\frac{\partial \pi_t^M}{\partial e_t} = \begin{cases} 0 & \text{if } t \text{ is non benchmark year} \\ \delta P_{1t+1} \left[\frac{\partial \pi_{t+1}^M}{\partial e_{t+1}} \mathbb{1}(\text{PY}_t \neq 3) + \frac{1}{3} \frac{\partial \pi_{t+1}^M}{\partial b_{t+1}} (1 - \lambda_{t+1}) \mathbb{1}(\text{PY}_t = 3) \right] & \text{otherwise} \end{cases}$$

where $P_{1t+1} = \Pr(d_{t+1} = 1 \mid \tilde{y}_{t+1}, b_{t+1})$. [▶ back](#)

State Variables Evolution

► Rolling Average Spending $\tilde{y}_t$

$$\tilde{y}_{it+1} = \begin{cases} y_{it} & \text{if } PY_t = 1 \\ \frac{1}{2}\tilde{y}_{it} \cdot \tilde{uf}_{it} + \frac{1}{2}y_{it} & \text{if } PY_t = 2 \\ \frac{2}{3}\tilde{y}_{it} \cdot \tilde{uf}_{it} + \frac{1}{3}y_{it} & \text{if } PY_t = 3 \end{cases}$$

The value of $\tilde{y}_{it+1}$ in the 3rd PY of each AP is used to rebase the benchmark for the subsequent AP.

► Updated Historical Regionalized Benchmark b_{it+1}

$$b_{it+1} = \begin{cases} [(1 - \lambda_{it+1})\tilde{y}_{it+1} + \lambda_{it+1}y_{it}^R] \cdot uf_{t+1}^1 & \text{if } PY_{t+1} = 1 \\ b_{it} \cdot uf_{t+1}^2 & \text{if } PY_{t+1} = 2 \\ b_{it} \cdot uf_{t+1}^3 & \text{if } PY_{t+1} = 3 \end{cases}$$

where uf_{t+1}^j is the updated factor applied to set the historical benchmark in PY j terms. [►► back](#)

Expectation Maximization

► Expectation Step:

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- (ii) Update the population probability of each type, w_k
- (iii) Using $\theta^{(m)}$ and $p^{(m)}$, update the CCP

$$p_1^{(m+1)} = \frac{1}{1 + \exp \left[v_{it}^F(\Omega_{it}, k; p^{(m)}, \theta^{(m)}) - v_{it}^M(\Omega_{it}, k; p^{(m)}, \theta^{(m)}) \right]}$$

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► Maximization Step: ► simulated value function

$$\theta^{(m+1)} = \operatorname{argmax}_{\theta} \sum_i \sum_k q_{ik} \log L_{ik}(\theta | q_{ik}^{(m+1)}, p_1^{(m+1)}, \Omega_{it}, k))$$

Simulated Value Function

To evaluate v_{it}^M , we use forward simulation to simulate S paths of participation effort, spending and profits

$$v_{it}^M \approx \frac{1}{S} \sum_{s=1}^S \sum_{h=0}^H \delta^h \pi_{i,t+h}^{(s)}$$

Simulated Value Function

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Given some value of the parameters θ , each simulated path uses:

- ▶ d_{it} draws from Kernel probabilities $\hat{Pr}(d_{it} = 1 | \Omega_{it})$
- ▶ e_{it} from the Euler equation [▶ euler equation](#)
- ▶ y_{it} from the spending equation
- ▶ $\tilde{y}_{it}$ and b_{it} move forward according to the MSSP rules. [▶ back](#)

Identification of variable costs

- ▶ From the model assumptions $y_{it}|e_{it} \sim N\left(\mathbb{E}y_{it}^{FFS} - e_{it}, \sigma_{it}\right)$

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- ▶ The Normal likelihood ties variation in Ω_{it} and x_{it} to the observed Δy_{it} , which identifies $(\beta_\gamma, \alpha_\gamma)$. ▶ FC

Identification of Fixed Costs

- ▶ Let Δv_t be the value difference between stay and exit net of the fixed cost:

$$\Delta v_t \equiv \tilde{v}_t^{\text{stay}}(\gamma; S_t) - v_t^{\text{exit}}$$

- ▶ Under logit, the log(Odds Ratio) is

$$\log \frac{P_t(\text{stay})}{1 - P_t(\text{stay})} = \underbrace{\Delta v_t}_{\text{offset}} - \underbrace{G_t}_{\text{horizon}} \cdot F$$

where G_t is the expected discounted number of future active periods if stay.

- ▶ We choose F so that the model-implied probabilities $P_t(\text{stay})$ fit the observed data best (maximum likelihood). [▶ back](#)

Likelihood Function

- ▶ Let $\theta = (\beta_C, \alpha_C, \beta_F, \alpha_F, \rho)$.

Likelihood Function

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- ▶ The likelihood function for ACO i :

$$L_i(\theta) = \sum_k q_{ik} \left[\prod_{t=1}^{T_i-1} \phi(y_{it}, \Omega_{it}, k; \theta) \Pr(d_{it} = 1 | \Omega_{it}, k; \theta) \right] \times \Pr(d_{iT_i} = 0 | \Omega_{iT_i}, k; \theta)$$

where q_{ik} s are the unobserved type probabilities , and

$$\phi(y_{it} | \Omega_{it}, k) = \frac{1}{\sqrt{2\pi\sigma_{it}^2}} \exp\left(-\frac{(y_{it} - \mu_{it} - e_{it})^2}{2\sigma_{it}^2}\right), \quad \mu_{it} = y_{it-1}^{FFS} \cdot \eta_{it} \cdot rr_{it}$$

and

$$\Pr(d_{it} = 1 | \Omega_{it}, k; \theta) = \frac{1}{1 + \exp(v_{it}^F - v_{it}^M)}$$

▶▶ back

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and

$$\Pr(d_{it} = 1 | \Omega_{it}, k; \theta) = \frac{1}{1 + \exp(v_{it}^F - v_{it}^M)}$$

▶ back

- ▶ I estimate θ with an EM algorithm (Arcidiacono Miller 2011).

▶ EM

Estimation Results: Variable Costs

	K=1	K=2	K=3	K=4
Risk Score	0.541** (0.156)	0.493*** (0.137)	0.435* (0.053)	0.451*** (0.061)
Beneficiaries	-0.241*** (0.059)	-0.193*** (0.039)	0.135* (0.035)	0.173*** (0.040)
Hospital	1.641** (0.316)	1.523** (0.358)	1.735** (0.415)	1.613** (0.421)
Type 1		0.302*** (0.045)		
Type 2		0.217*** (0.052)	0.134** (0.043)	
Type 3		0.147** (0.042)	0.094** (0.028)	0.056 (0.053)
GOF	0.852	0.881	0.936	0.904

(i) Coefficients represent the dollar change in the cost of reducing spending by 1%. (ii) Standard errors are computed with bootstrapping method

Estimation Results: Fixed Costs

	K=1	K=2	K=3	K=4
Beneficiaries	0.142** (0.059)	0.194** (0.039)	0.136** (0.042)	0.154** (0.041)
Hospital	0.640** (0.116)	0.554** (0.158)	0.637** (0.135)	0.643** (0.121)
Type 1		0.252*** (0.045)		
Type 2		0.253*** (0.052)	0.216*** (0.043)	
Type 3		0.247** (0.068)	0.214** (0.057)	0.156* (0.083)
GOF	0.852	0.881	0.936	0.904

(i) Coefficients represent the dollar change in the cost of reducing spending by 1%. (ii) Standard errors are computed with bootstrapping method

Benchmark under Conditional Regionalization

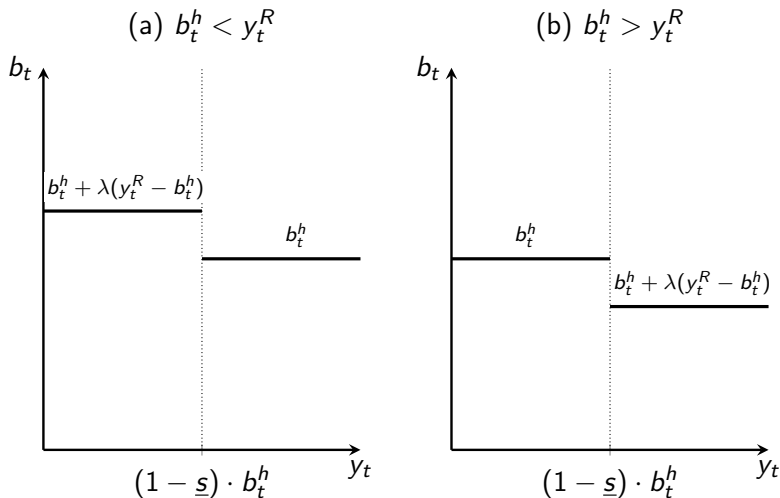


Figure 3: Conditional Regionalization

Counterfactual Results

Table 3: COUNTERFACTUALS
(Percentage change relative to status quo)

	Part Rate P	% Δ Spending Gross SS Δy	SS Payments Δ_{ss}	% Δ Spending Net of SS ΔY
Simul MSSP	56.2	0	0	0
No Regn	46.4	0.13	-0.71	-0.58
No Rebas	65.5	-2.12	0.58	-1.54
Cond Regn	58.7	-2.86	0.73	-2.13

Notes: (1) P is the MSSP participation rate, (2) Δy is the % change in spending per-capita among all ACOs, (3) Δ_{ss} is the % change in Shared Savings Payments, and (4) ΔY is the % change in spending per-capita net of Shared Savings Payments among ACOs.

Counterfactual Results

Table 4: COUNTERFACTUALS
(Percentage change relative to status quo)

	Part Rate P	% Δ Spending Gross SS Δy	SS Payments Δ_{ss}	% Δ Spending Net of SS ΔY
Simul MSSP	56.2	0	0	0
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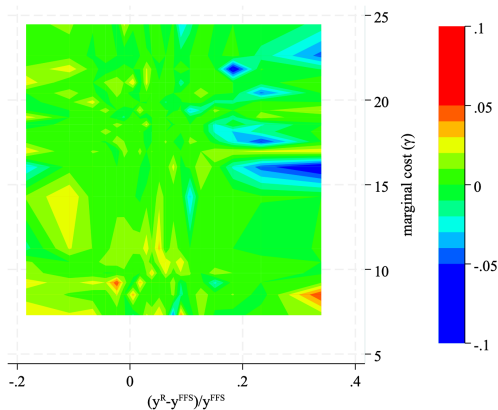
Counterfactual Results

Table 5: COUNTERFACTUALS
(Percentage change relative to status quo)

	Part Rate P	% Δ Spending Gross SS Δy	SS Payments Δ_{ss}	% Δ Spending Net of SS ΔY
Simul MSSP	56.2	0	0	0
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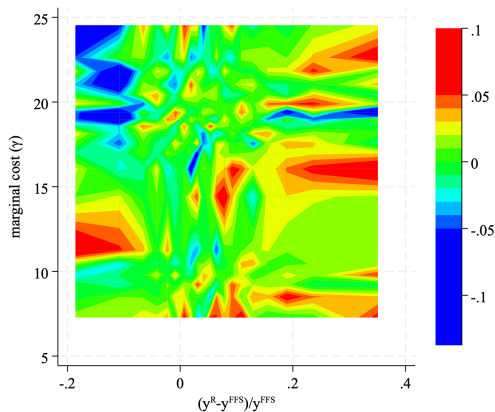
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Savings under No Regionalization



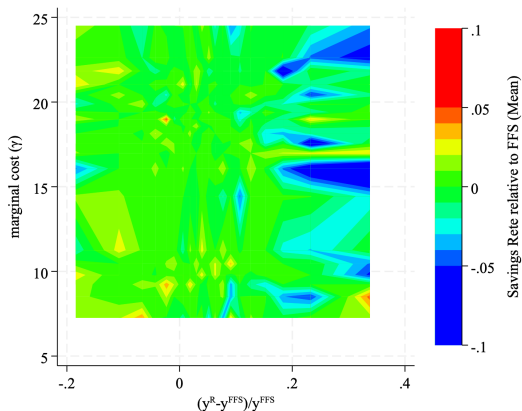
$$\text{sav rate relative to FFS} = \frac{y^{FFS} - y^{MSSP} - SS}{y^{FFS}}$$

Savings under No Rebasement



$$\text{sav rate relative to FFS} = \frac{y^{FFS} - y^{MSSP} - SS}{y^{FFS}}$$

Savings under Conditional Regionalization



$$\text{sav rate relative to FFS} = \frac{y^{FFS} - y^{MSSP} - SS}{y^{FFS}}$$